

TIC 86934109

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.052257 d
Epoch (t0) 1684.0610 BTJD
Duration 0.5000 d
Depth 0.457 % (4,569 ppm)
SNR 111.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

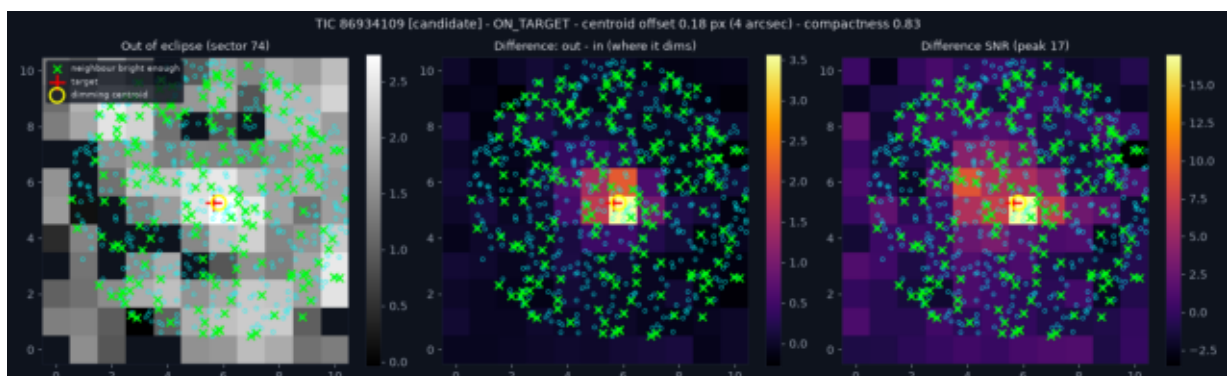
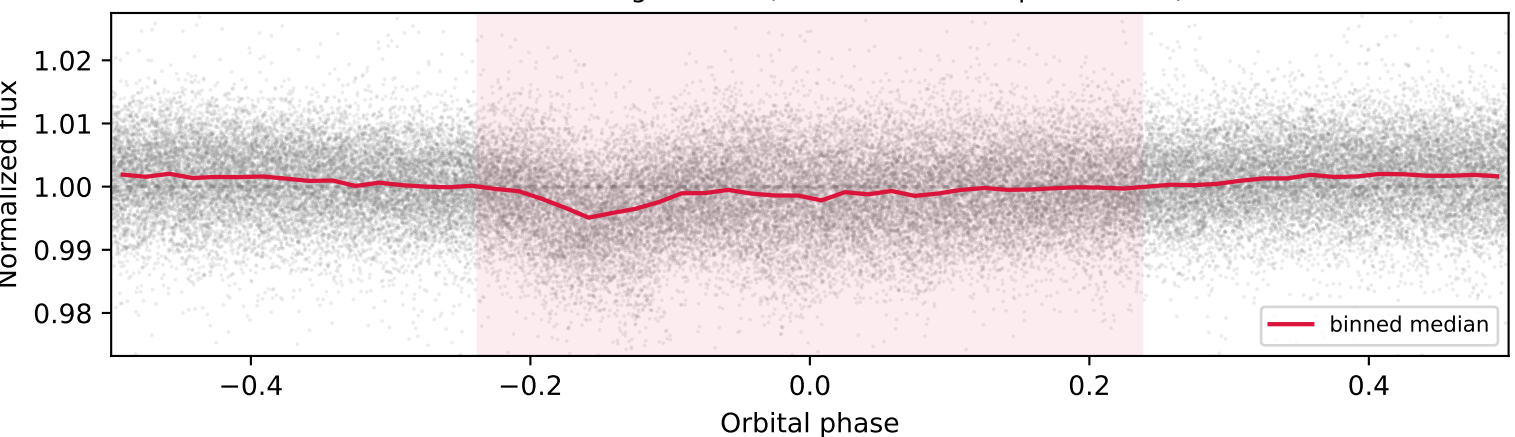
Eclipses observed 204 (gate ≥ 3)
Robust depth ratio 0.53 (gate ≥ 0.50)
Robust depth sigma 54.6 (gate ≥ 7.0)
Odd/even z-score 0.73 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 74 light-curve dip in this sector: 13.2 sigma t0 refined by -0.1975 phase
Difference-image peak SNR 17.04 (gate ≥ 5.0) Compactness 0.83 (gate ≥ 0.50)
Dimming centroid offset 0.18 px (4 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 86934145
0.65 px away, dTmag +5.42 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.